Quadratic stabilization of the canonical Foulkes–Howe map

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Abstract

For every finite-dimensional complex vector space V, we prove that the canonical Foulkes–Howe map $\mathrm{Sym}^b(\mathrm{Sym}^a V) \longrightarrow \mathrm{Sym}^a(\mathrm{Sym}^b V)$ is surjective whenever $a \geq 2$ and $b \geq a(a-1)$. This gives a quadratic stabilization bound independent of $\dim V$.

1 Introduction

Let V be a finite-dimensional complex vector space. Products of vectors will always mean products in the symmetric algebra $\mathrm{Sym}\, V$. For an integer $a \geq 1$ and each integer $j \geq 0$, put

$$P_a = (\mathrm{Sym}\, V)^{\otimes a}, \qquad R_j = \big((\mathrm{Sym}^j V)^{\otimes a}\big)^{\mathfrak{S}_a}, \qquad R = \bigoplus_{j \geq 0} R_j.$$

The group $\mathfrak{S}_a$ permutes the tensor factors, and multiplication in P_a makes R a graded commutative algebra. We use the averaging isomorphism from a symmetric power to the corresponding invariant tensors: for any complex vector space W, it sends $w_1 \cdots w_a \in \mathrm{Sym}^a W$ to

$$\mathrm{Av}(w_1 \otimes \cdots \otimes w_a) = \frac{1}{a!} \sum_{\pi \in \mathfrak{S}_a} w_{\pi(1)} \otimes \cdots \otimes w_{\pi(a)}. \tag{1}$$

This map is well defined on the symmetric-power quotient, and the quotient projection restricted to invariant tensors is its inverse. In particular, $R_1 = \mathrm{Sym}^a V$ and $R_b = \mathrm{Sym}^a(\mathrm{Sym}^b V)$ under these identifications. For every integer $b \geq 0$, multiplication in P_a induces a map $\mathrm{Sym}^b R_1 \longrightarrow R_b$. Under the averaging identifications, this is the *canonical Foulkes–Howe map* considered here:

$$\mu_{a,b,V} : \mathrm{Sym}^b(\mathrm{Sym}^a V) \longrightarrow \mathrm{Sym}^a(\mathrm{Sym}^b V). \tag{2}$$

If $A \subseteq R$ is the subalgebra generated by R_1, then $\mathrm{im}\, \mu_{a,b,V} = A_b$. In positive dimension, Raicu–Sam–Weyman [11, Section 2] formulate this map as $\mathrm{Sym}^b(D^a V) \longrightarrow D^a(\mathrm{Sym}^b V)$, with divided powers realized as invariant tensors. Transporting both sides by the averaging above gives exactly (2).

For fixed a and V, the stabilization question asks for a nonnegative degree b_0 such that $\mu_{a,b,V}$ is surjective for every $b \geq b_0$. One may ask for the least such degree in a fixed dimension, or for a bound valid in every finite dimension at once. We prove the following uniform sufficient bound.

Theorem 1. *For every integer $a \geq 2$, every integer $b \geq a(a-1)$, and every finite-dimensional complex vector space V, the map $\mu_{a,b,V}$ in (2) is surjective. Equivalently, $A_b = R_b$.*

The case $a = 1$ is the identity map in every degree.

History and significance. The affine Chow variety of decomposable a-forms consists of products of a linear forms in $\mathrm{Sym}^a(V^*)$. Its coordinate ring identifies with A, whose normalization is R; here b is the coordinate degree [11, Section 2, Theorem 2.3]. This interpretation connects the canonical map to the geometry of symmetric products; see also Landsberg [8, Section 7.6, Propositions 7.11–7.12]. Brion established eventual surjectivity [2] and subsequently gave an effective bound involving both a and the dimension [3, Theorem 3.3]. In the normalization picture, eventual surjectivity says that the graded quotient R/A has only finitely many nonzero components. The least stabilization degree is one more than its highest nonzero degree, or zero if $R = A$. Raicu, Sam, and Weyman bound the regularity of the Segre ring and modules of covariants, including the normalization [11, Theorem 3.9]. Their theorem does not compute the top degree of the separate quotient R/A that controls stabilization. Theorem 1 gives the sufficient bound $a(a-1)$ in every dimension. It also answers positively the request for a polynomial bound in [8, Problem 7.19]: in that source's notation $d = b$, $n = a$, and $w = \dim V$, the bound is $d \geq n(n-1)$, including when $w = n$. The present argument does not determine the least stabilization degree in general; the bound is not asserted to be sharp.

The canonical-map question must be distinguished from the ordinary Foulkes conjecture [5], which asks, for $a \leq b$, for a $\mathrm{GL}(V)$-equivariant injection $\mathrm{Sym}^a(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V)$ [11, Conjecture 2.10]. Surjectivity in Theorem 1 implies that representation comparison in the stated range: complete reducibility over $\mathbb{C}$ splits the surjection and supplies an equivariant embedding in the opposite direction. Corollary 4 gives the precise statement. Existence of an embedding alone does not specify the canonical map. At $a = b$ the two representations are the same, so their ordinary comparison is automatic, while a specified canonical endomorphism can still fail to be invertible. Indeed, the canonical map has nonzero kernels at the diagonal pairs $(a, b) = (5, 5)$ and $(6, 6)$ in suitable dimensions [4, arXiv version, Theorem 6(b),(c) and Remark 7], both outside the range of Theorem 1.

The proof mechanism. An annihilator criterion reduces Theorem 1 to a vanishing statement for symmetric a-linear forms on $\mathrm{Sym}^b V$ whose diagonal vanishes on products of b vectors. Factoring a binary form first gives an identity with a common product in every slot. Each of two differential sequences then starts from an evaluation whose vanishing is sought and inserts factors until it reaches an identity already known to be zero. Injectivity recovers vanishing before those insertions. The first sequence frees the last slot from the common product; induction lets the product factors in the other slots vary independently; and the second sequence removes their remaining common power from the vanishing conclusion. A directional derivative in a vector variable U toward a second variable Y lowers the U-degree and raises the Y-degree by one. The differential lemma proves injectivity on a homogeneous space whenever the U-degree exceeds the Y-degree. The first sequence uses a different Y at each step; the second raises the degree of one shared Y at every step, and this accumulation leads to the quadratic sufficient bound.

Successive weight shifts also occur in Ikenmeyer's proof [7, Section 5] of McKay's propagation theorem [10]. In Ikenmeyer's formulation, the canonical maps in the opposite direction $\mathrm{Sym}^a(\mathrm{Sym}^b V) \longrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V)$ are considered with $\dim V \geq b$ for each b. For fixed a and $B \geq a$, injectivity at (a, B) propagates to every (a, b) with $b \geq B$ [7, Theorem 1.3 and Section 3]. Here the two sequences surround an induction on the number of slots and give a surjectivity range directly. Section 2 establishes the diagonal criterion, Section 3 proves the one-step injectivity statement, and Section 4 carries out the induction.

2 The canonical image and diagonal vanishing

We first express surjectivity of the specified multiplication map as a vanishing problem. Fix integers $a, b \geq 1$ and put $E = \operatorname{Sym}^b V$. An element $P = u_1 \cdots u_b \in E$ will be called a *product of b vectors*; the vectors may repeat or be zero.

For every integer $k \geq 1$, pure powers span $\operatorname{Sym}^k V$. Indeed, the polarization identity

$$k!\, u_1 \cdots u_k = \sum_{J \subseteq \{1,\dots,k\}} (-1)^{k-|J|} \left(\sum_{j \in J} u_j \right)^k$$

holds in the symmetric algebra. On expanding first with the u_i as formal generators, every monomial missing an index cancels, and the only degree-k monomial containing all k indices has coefficient $k!$. Substituting arbitrary vectors proves the identity. Products span $\operatorname{Sym}^k V$, and $k! \neq 0$ over $\mathbb{C}$, so pure powers span it as well.

Averaging sends $u^a \in \operatorname{Sym}^a V$ exactly to $z(u) = u^{\otimes a} \in R_1$. The spanning fact with $k = a$ therefore says that the elements $z(u)$ span R_1. Componentwise multiplication gives

$$z(u_1) \cdots z(u_b) = (u_1 \cdots u_b)^{\otimes a},$$

so multiplying spanning generators yields the exact image description

$$A_b = \operatorname{span}_{\mathbb{C}} \{ P^{\otimes a} : P \text{ is a product of } b \text{ vectors} \}.$$

For $\lambda \in R_b^*$, define the symmetric a-linear form on E

$$T_\lambda(w_1, \dots, w_a) = \lambda(\operatorname{Av}(w_1 \otimes \cdots \otimes w_a)).$$

Every symmetric a-linear form arises this way: its linearization on $E^{\otimes a}$ restricts to a functional on R_b, and symmetry makes averaging leave its value unchanged. The functional is unique because averages of simple tensors span R_b. For a product P, $T_\lambda(P, \dots, P) = \lambda(P^{\otimes a})$. Thus the annihilator of A_b corresponds exactly to the symmetric forms whose diagonal vanishes on every product of b vectors. Since R_b is finite-dimensional, $A_b = R_b$ if and only if that annihilator is zero. We have proved the following criterion:

The map $\mu_{a,b,V}$ is surjective if and only if every symmetric a-linear form T on $\operatorname{Sym}^b V$ satisfying $T(P, \dots, P) = 0$ for all products P of b vectors is zero.

This also explains the reversed roles of the two parameters in the equation interpretation. The multiplication algebra above concerns products of a linear forms in coordinate degree b. Here $T(P, \dots, P)$ is a homogeneous degree-a polynomial on $\operatorname{Sym}^b V$, tested on products of b vectors, viewed as linear forms on V^*. The pairing just proved connects these two descriptions; compare the product parametrization in [11, Section 2].

3 Differential injectivity

The induction will recover a polynomial from a sequence of its directional derivatives. We first give the injectivity criterion that permits each derivative to be cancelled.

Choose a basis of V, of dimension $n > 0$. By a vector variable we mean an independent block of n scalar coordinate variables. For two such blocks $U = (U_1, \dots, U_n)$ and $Y = (Y_1, \dots, Y_n)$, define

$$D_{Y,U} = \sum_{k=1}^{n} Y_k \frac{\partial}{\partial U_k}.$$

This operator lowers the U-degree by one and raises the Y-degree by one. The criterion below is the positive-weight injectivity in the usual $\mathfrak{sl}_2$ weight strings; compare [6, Section 11.1, Claim 11.4 and equation (11.5)]. Ikenmeyer's argument [7, Section 5, Claim 5.1] decomposes tensors according to untouched positions before applying a two-coordinate weight shift. We give a direct inner-product proof with all additional variables present.

Lemma 2. *Let p, q be nonnegative integers. On polynomials homogeneous of degree p in U and degree q in Y, the operator $D_{Y,U}$ is injective if $p > q$. The polynomials may depend on any finite collection of additional variables independent of U and Y.*

Proof. Give the polynomial ring the positive definite Fischer, or apolar, Hermitian inner product [1, Section 2, equation (8) and Proposition 3], in which distinct monomials are orthogonal and

$$\|X^\alpha\|^2 = \alpha! := \prod_k \alpha_k!;$$

here X lists all coordinates, including the additional variables. Differentiation by a coordinate is adjoint to multiplication by that coordinate. Thus, for $D = D_{Y,U}$ and $E = D_{U,Y}$, one has $D^* = E$. A direct calculation gives

$$ED - DE = N_U - N_Y, \qquad N_U = \sum_k U_k \partial_{U_k}, \quad N_Y = \sum_k Y_k \partial_{Y_k}.$$

Consequently, for a polynomial h of the indicated bidegree,

$$\|Dh\|^2 - \|Eh\|^2 = (p - q)\|h\|^2. \tag{3}$$

If $Dh = 0$ and $p > q$, the left side is nonpositive and the right side is nonnegative, with equality only when $h = 0$. □

Every application of Lemma 2 below concerns the entire intermediate homogeneous space. Accordingly, a sequence of the operators is injective whenever its degree inequality holds before each individual operator.

4 Vanishing on products

We now prove the vanishing statement in the criterion of Section 2. Its formulation uses only symmetric multilinear forms and products in $\operatorname{Sym} V$.

Proposition 3. *Let V be a finite-dimensional complex vector space, and let $a \geq 1$ and $b \geq 1$ be integers satisfying $b \geq a(a-1)$. If a symmetric a-linear form T on $\operatorname{Sym}^b V$ satisfies*

$$T(P, \dots, P) = 0 \qquad \text{for every product } P \text{ of } b \text{ vectors}, \tag{4}$$

then $T = 0$.

Proof. The assertion is immediate for $V = 0$, so suppose $\dim V > 0$. We induct on a, proving the assertion simultaneously for every admissible b and every finite-dimensional V. For $a = 1$, products span $\operatorname{Sym}^b V$. Now let $a \geq 2$, assume the assertion for $a - 1$, and put

$$r = a - 1, \qquad m = b - r. \tag{5}$$

Our hypotheses give

$$b \geq r(r+1), \qquad m \geq r^2 \geq 1. \tag{6}$$

Step 1: a split binary form. Let $Q = y_1 \cdots y_m$ be a product of m vectors. For every $c \in \mathbb{C}$ the homogeneous binary polynomial $X^r + cY^r$ splits into r linear factors over $\mathbb{C}$: for $c \neq 0$, choose $\alpha \in \mathbb{C}$ with $\alpha^r = -c$ and $\zeta = e^{2\pi i/r}$, so that

$$X^r + cY^r = \prod_{j=0}^{r-1} (X - \alpha\zeta^j Y).$$

For $c = 0$ the factorization is X^r. Substitution in $\operatorname{Sym} V$ therefore shows, without requiring x and t to be independent, that

$$P = (x^r + ct^r)Q$$

is a product of b vectors for all $x, t \in V$. For fixed x, t, Q, condition (4) says that the polynomial $T(P, \dots, P)$ in c vanishes identically. By symmetry and multilinearity, its coefficient of c is $a\, T(x^rQ, \dots, x^rQ, t^rQ)$. Hence

$$T(\underbrace{x^rQ, \dots, x^rQ}_{r \text{ slots}}, t^rQ) = 0. \tag{7}$$

Step 2: freeing the last slot. We seek the same vanishing with the last input replaced by the independent pure power t^b. Differentiation will insert the missing factors of Q into that input, producing (7); injectivity will then recover the vanishing before those insertions. Fix $x \in V$, regard $t, y_1, \dots, y_m$ as independent vector variables, and set

$$H_x = T(\underbrace{x^rQ, \dots, x^rQ}_{r \text{ slots}}, t^b).$$

It is homogeneous of degree b in t and degree r in each y_i. Differentiation in t acts only on the last slot. Thus, for $0 \leq k \leq m$,

$$D_{y_k,t} \cdots D_{y_1,t} H_x = \frac{b!}{(b-k)!} T(x^rQ, \dots, x^rQ, t^{b-k} y_1 \cdots y_k), \tag{8}$$

with the empty composition understood when $k = 0$. At $k = m$ the right side is zero by (7), first for every choice of vector values and hence as a polynomial. Here and below we use the elementary fact that a polynomial in finitely many variables over $\mathbb{C}$ vanishing at every point is zero; it follows by induction on the number of variables from the univariate root bound. Immediately before the i-th operator, the degrees in t and y_i are

$$p = b - i + 1, \qquad q = r, \qquad p - q = m - i + 1 \geq 1.$$

Here y_i already has degree r in H_x, but no earlier operator has increased its degree. Lemma 2 makes every operator injective, and hence $H_x = 0$.

Step 3: induction on the remaining slots. For fixed $x, t \in V$, the expression

$$S_{x,t}(Q_1, \dots, Q_r) := T(x^rQ_1, \dots, x^rQ_r, t^b)$$

is a symmetric r-linear form on $\operatorname{Sym}^m V$. Step 2 says that its diagonal vanishes at every product of m vectors. Since $m \geq r^2 \geq r(r-1)$ and $m \geq 1$, the induction hypothesis applies and gives

$$T(x^rQ_1, \dots, x^rQ_r, t^b) = 0 \qquad (Q_1, \dots, Q_r \in \operatorname{Sym}^m V). \tag{9}$$

This identity holds for every $x, t \in V$.

Step 4: removing the common power. The factors Q_i in (9) may now be chosen independently; their remaining shared constraint is the multiplier x^r. We will differentiate evaluations on separately chosen pure powers into this known zero and again use injectivity to recover their vanishing. Fix $t \in V$ and form the polynomial

$$f = T(v_1^b, \dots, v_r^b, t^b)$$

in independent vector variables $v_1, \dots, v_r$. Introduce one further independent vector variable x; initially f has x-degree zero. Apply D^r_{x,v_i} successively for $i = 1, \dots, r$. Each derivative acts only on its own v_i variable, so it does not differentiate the previously inserted x factors or any other slot. The combined result is

$$D^r_{x,v_r} \cdots D^r_{x,v_1} f = \left(\frac{b!}{m!}\right)^r T(x^r v_1^m, \dots, x^r v_r^m, t^b) = 0. \tag{10}$$

The last equality follows from (9) for every choice of the vector values, hence as a polynomial identity over $\mathbb{C}$.

All r^2 individual operators raise the degree of the same variable block x. Before any one is applied, its degree is therefore at most $r^2 - 1$. The source v_i of that operator has previously been differentiated at most $r-1$ times, so its degree is at least $b-r+1 = m+1 \geq r^2+1$. Thus the source degree exceeds the destination degree before every operator, with all other blocks as spectators. Lemma 2 makes the composition injective, and (10) implies $f = 0$. As t is arbitrary, T vanishes on every tuple of pure powers.

Pure powers span $\mathrm{Sym}^b V$ by the polarization argument in Section 2. Multilinearity now yields $T = 0$, completing the induction. □

Proof of Theorem 1. For $a \geq 2$ and $b \geq a(a-1)$, Proposition 3 supplies the vanishing assertion in the criterion of Section 2. That criterion gives the surjectivity of $\mu_{a,b,V}$. □

5 Representation comparison and scope

Corollary 4. *For every finite-dimensional complex vector space V, every integer $a \geq 2$, and every integer $b \geq a(a-1)$, there exists a $\mathrm{GL}(V)$-equivariant embedding*

$$\mathrm{Sym}^a(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V).$$

In particular, this gives the sixth symmetric-power comparison for $b \geq 30$.

Proof. Averaging and multiplication commute with the action of $\mathrm{GL}(V)$, so the surjection $\mu_{a,b,V}$ of Theorem 1 is equivariant. Finite-dimensional polynomial representations over $\mathbb{C}$ are completely reducible [9, Chapter I, Appendix A, Sections 7–8]. Choose an invariant complement M to its kernel. The restriction $\mu_{a,b,V}|_M$ is an equivariant isomorphism onto $\mathrm{Sym}^a(\mathrm{Sym}^b V)$. Its inverse followed by the inclusion of M in $\mathrm{Sym}^b(\mathrm{Sym}^a V)$ is the required embedding. This construction asserts existence and does not identify the embedding with a prescribed canonical map. □

Remark 5. The differential lemma holds for every finite number of coordinates and arbitrary additional variables, while the degree comparisons depend only on a and b. Together with the finite-dimensional annihilator argument, this proves the same bound in every finite dimension, without a stable-dimension reduction. The written proof uses positivity of the Fischer Hermitian inner product, splitting of binary forms over $\mathbb{C}$, and the nonzero characteristic-zero scalar factors in the coefficient and differentiation identities.

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